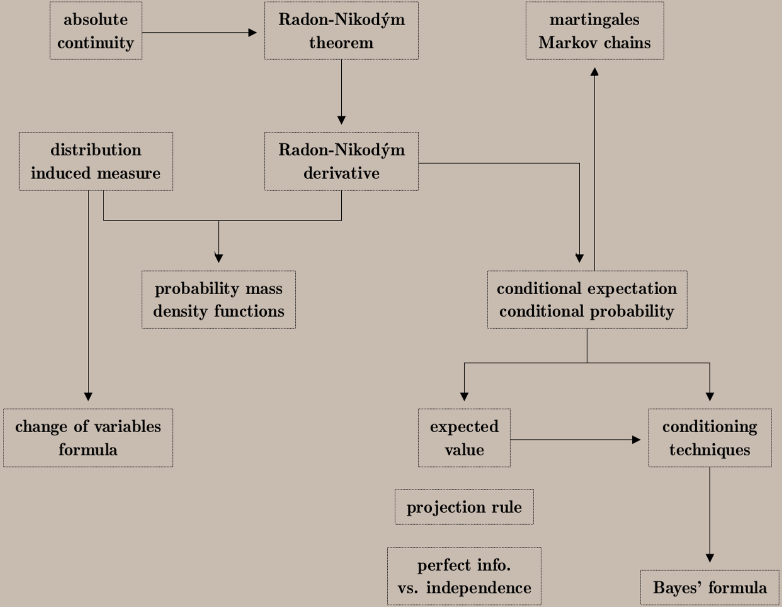


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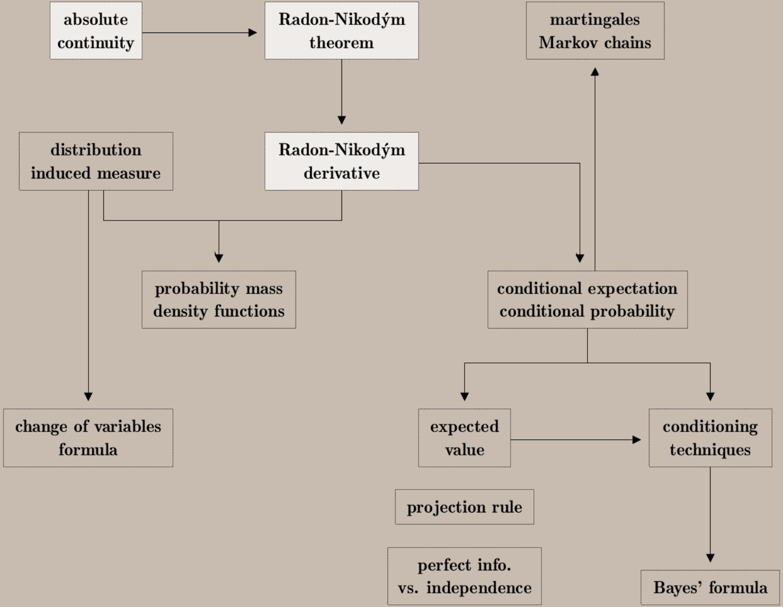

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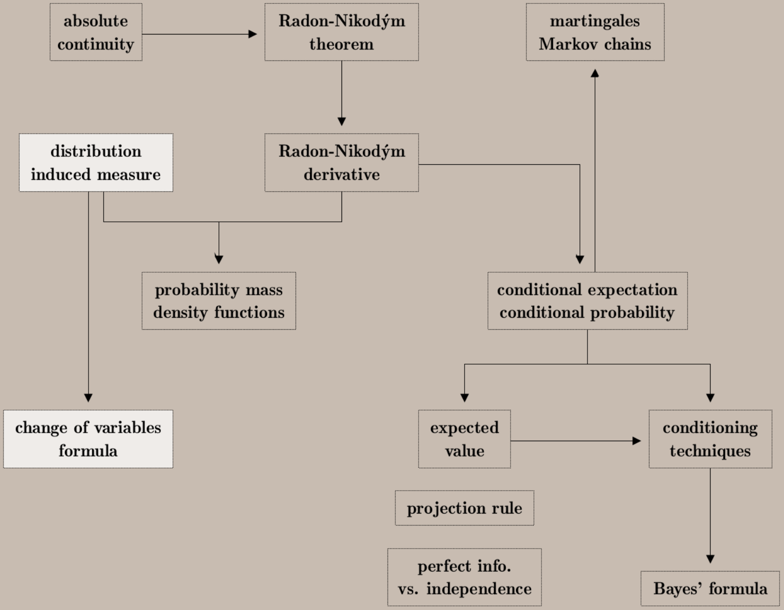

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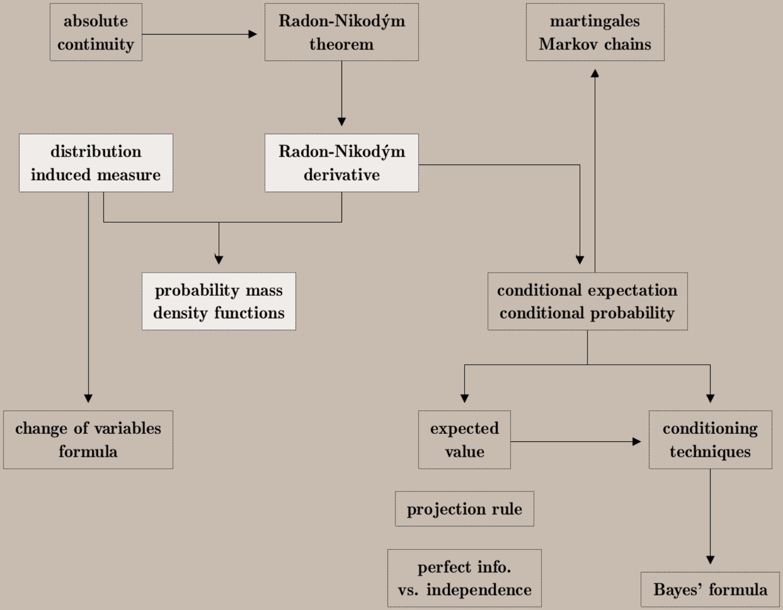

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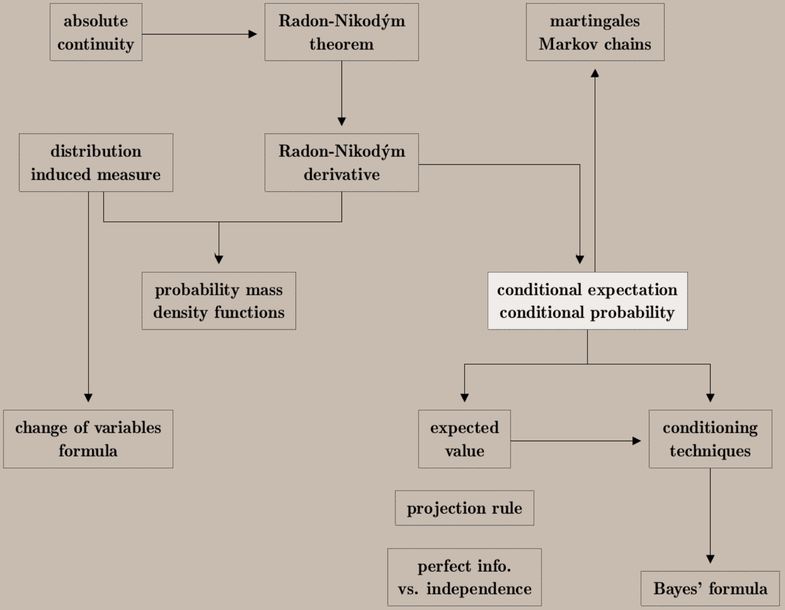


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absolute continuity

Radon-Nikodým theorem

martingales
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probability mass
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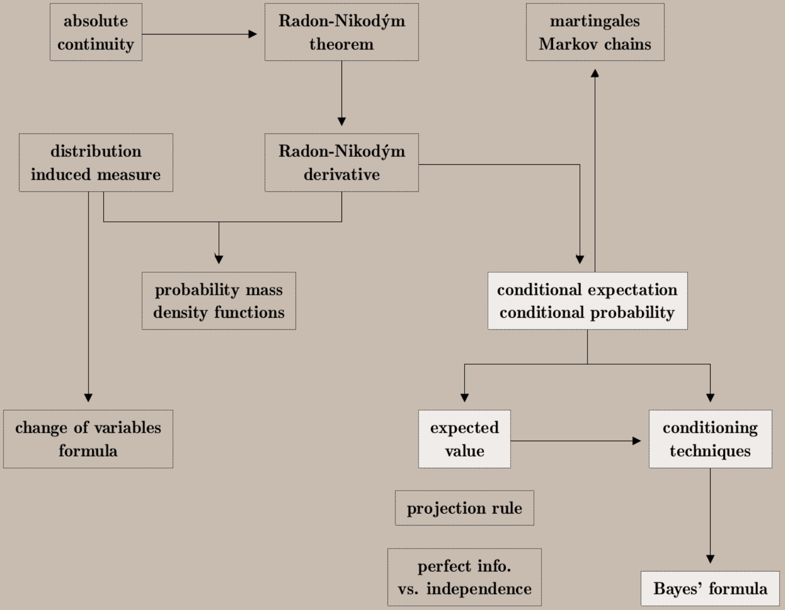

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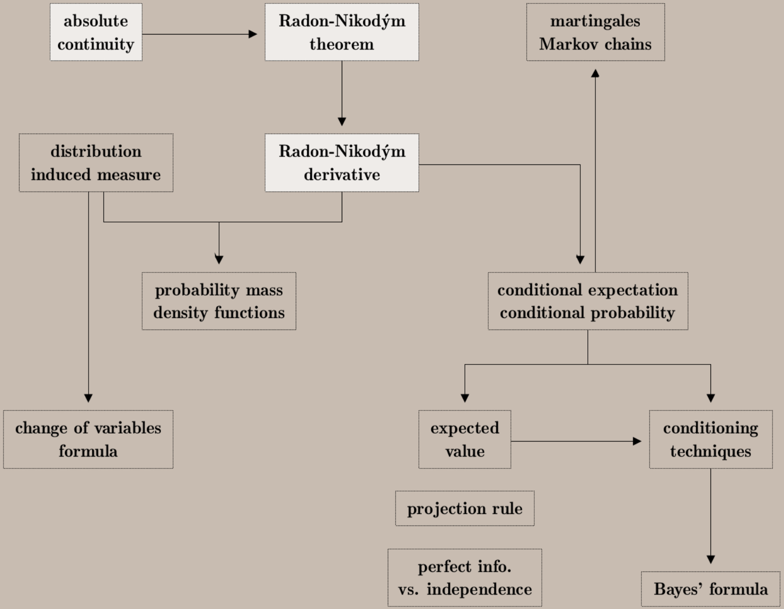


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Otto Nikodým

Motivation: Let μ = positive measure, and $\phi \geq 0$ be measurable.

$$\nu(A) = \int_A \phi \, d\mu = \int \phi \, \mathbf{1}_A \, d\mu \quad \text{for all} \quad A \in \mathscr{F} \quad \text{is a positive measure.}$$

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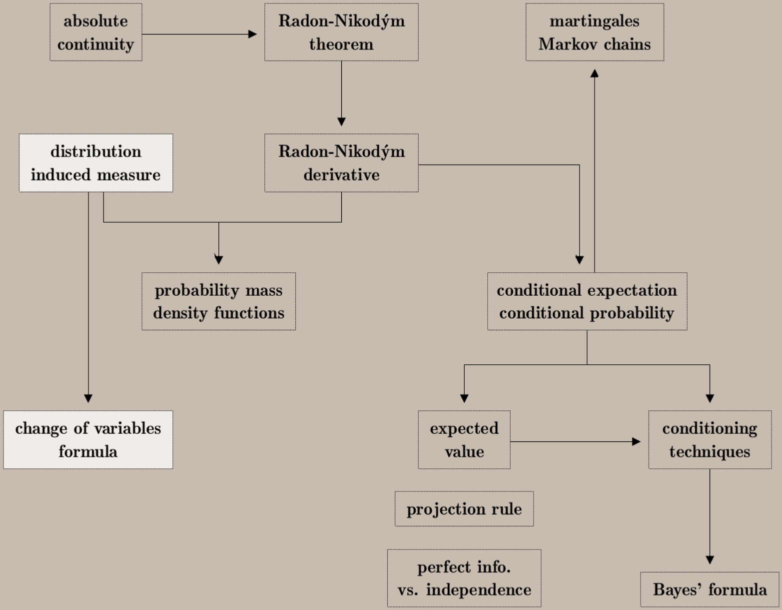

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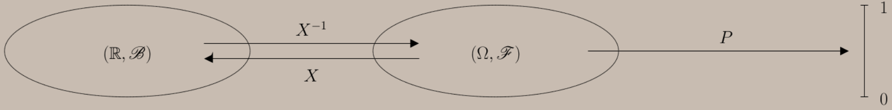

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h \text{ integrable} \quad \Longrightarrow \quad E(h(X)) &= E(h^+(X) - h^-(X)) \overset{\text{LIN}}{=} E(h^+(X)) - E(h^-(X)) \\
&\overset{\text{S3}}{=} \int h^+\, d\nu_X - \int h^-\, d\nu_X \overset{\text{LIN}}{=} \int h\, d\nu_X.
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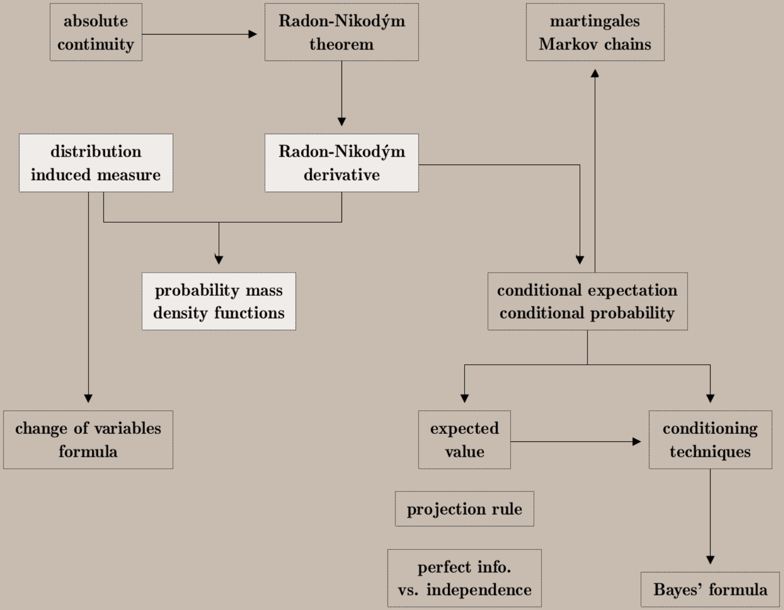

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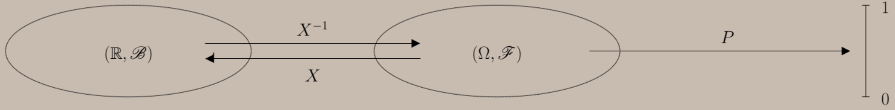

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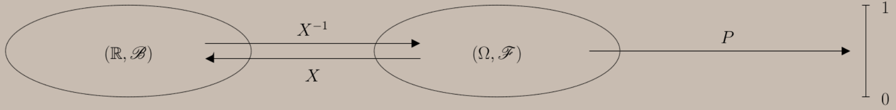

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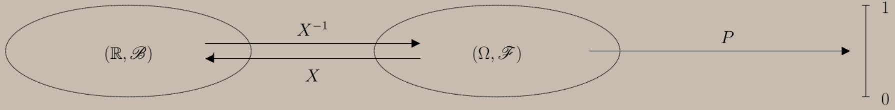

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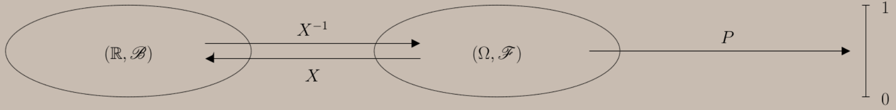

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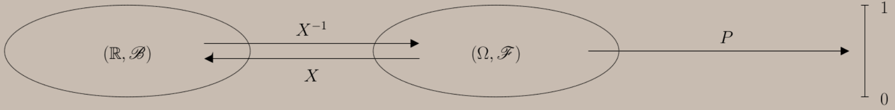

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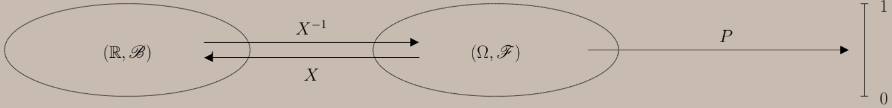

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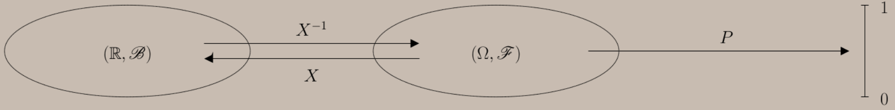

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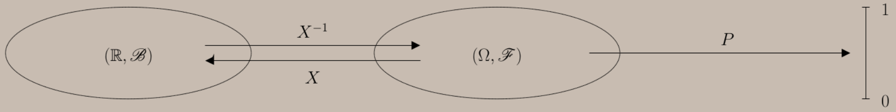
Distribution: $\nu_X(B) = P(X \in B) = \int \mathbf{1}_{X^{-1}(B)}\, dP$ for all $B \in \mathscr{B}$.



Change of variables formula:

$$E(h(X)) = \int_\Omega h(X)\, dP = \int_\mathbb{R} h\, d\nu_X \quad \text{for all } h \text{ positive or integrable.}$$

Continuous random variable: $\nu_X \ll \lambda =$ Lebesgue measure $\lambda([a,b]) = b - a$.

RNT $\Longrightarrow \phi_X = d\nu_X / d\lambda =$ probability density function of X

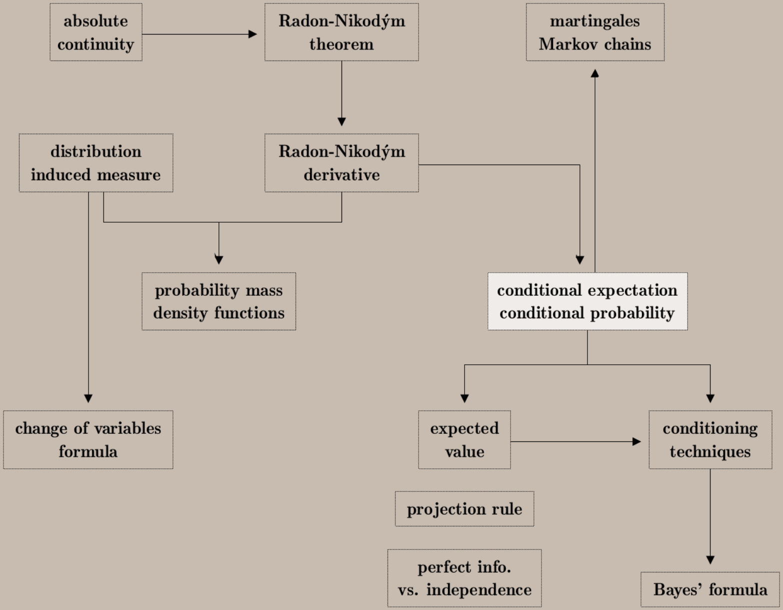
$\stackrel{\text{CVF}}{\Longrightarrow} E(h(X)) = \int h\, d\nu_X = \int h(x)\, \phi_X(x)\, \lambda(dx)$.

Discrete random variable: $\nu_X \ll \mu_\mathbb{N} =$ counting measure $\mu_\mathbb{N}(A) = \text{card}(A)$ for all $A \subset \mathbb{N}$.

RNT $\Longrightarrow \phi_X = d\nu_X / d\mu_\mathbb{N} =$ probability mass function of X

$\stackrel{\text{CVF}}{\Longrightarrow} E(h(X)) = \int h\, d\nu_X = \sum_x h(x)\, \phi_X(x)$.

Conclusion: the Radon–Nikodým derivative unifies the discrete and the continuous cases.



Johann Radon

Otto Nikodým

Definition. Let $X \in L^1(\Omega, \mathscr{F}, P)$, and $\mathscr{G} \subset \mathscr{F}$ be a sub-σ-algebra.

The conditional expectation of X given $\mathscr{G}$ is any random variable $Z = E(X \,|\, \mathscr{G})$ such that

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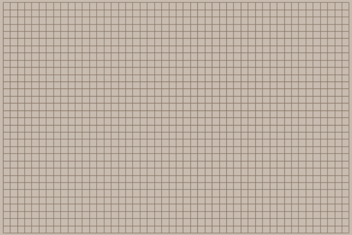
Random variable X

Sub-σ-algebra $\mathscr{G} \subset \mathscr{F}$



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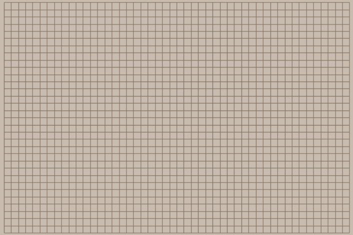

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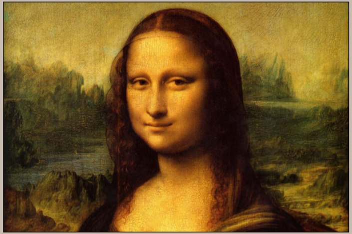
   Same color across each given square. The larger the σ-algebra, the better the resolution.



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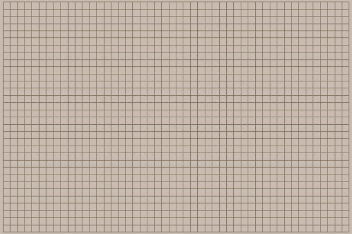

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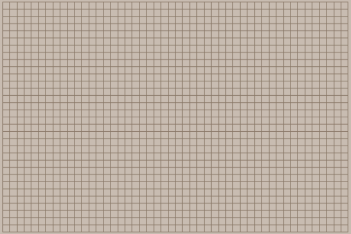

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Conditional expectation $E(X \,|\, \mathscr{G})$

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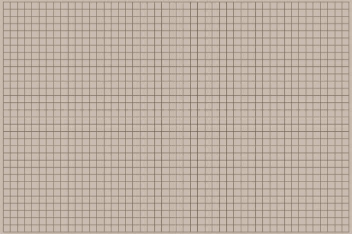
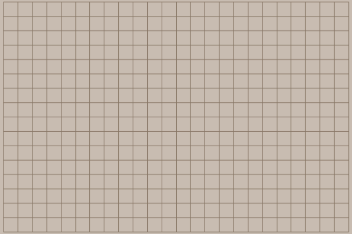


Smaller σ-algebra $\mathscr{H} \subset \mathscr{G}$



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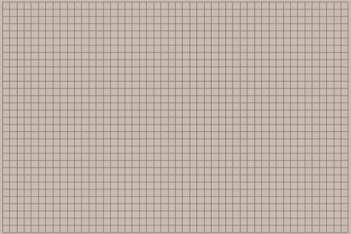

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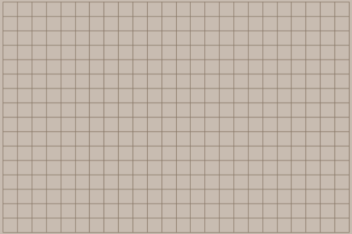

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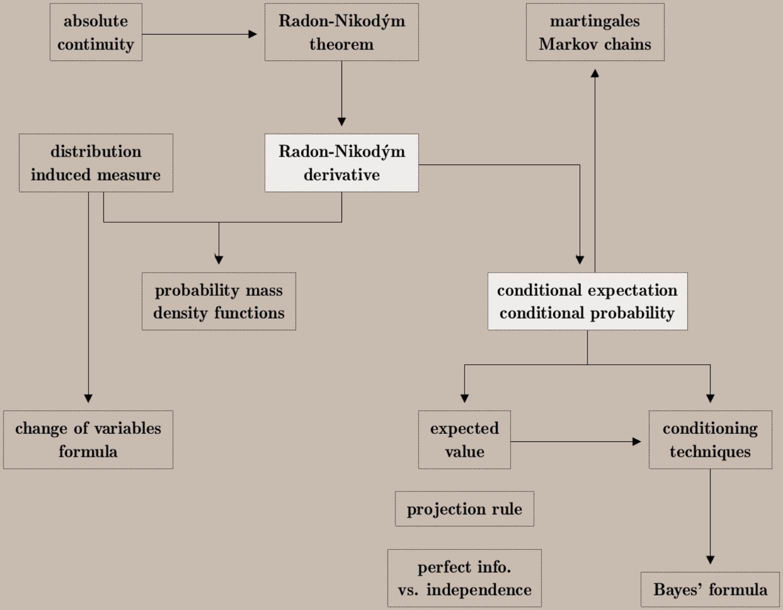


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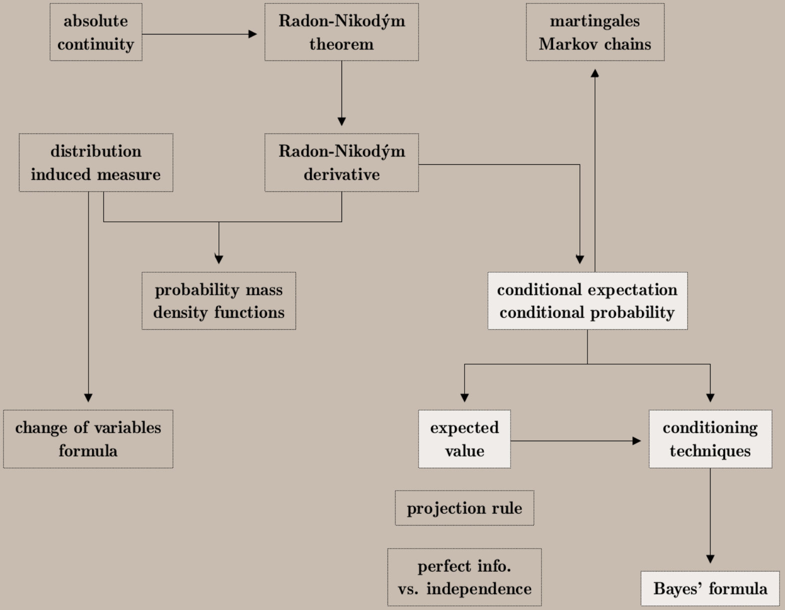
Assume that X is integrable.

$$E(X\mathbf{1}_A) \overset{\text{LIN}}{=} E(X^+\mathbf{1}_A) - E(X^-\mathbf{1}_A) \overset{\text{S1}}{=} E(Z^+\mathbf{1}_A) - E(Z^-\mathbf{1}_A) \overset{\text{LIN}}{=} E(Z\mathbf{1}_A)$$

for all $A \in \mathscr{G}$ where $Z = Z^+ - Z^-$ is $\mathscr{G}$-measurable.



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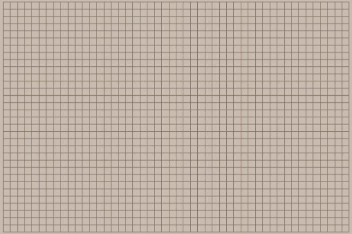

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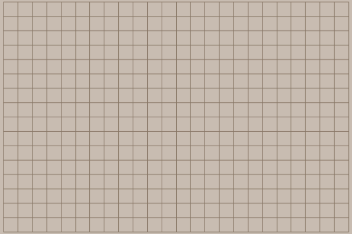

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Conditional expectation $E(X \mid \mathscr{G})$

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Conditioning on a random variable: Take $\mathscr{G} = \sigma(Y)$ with Y discrete.

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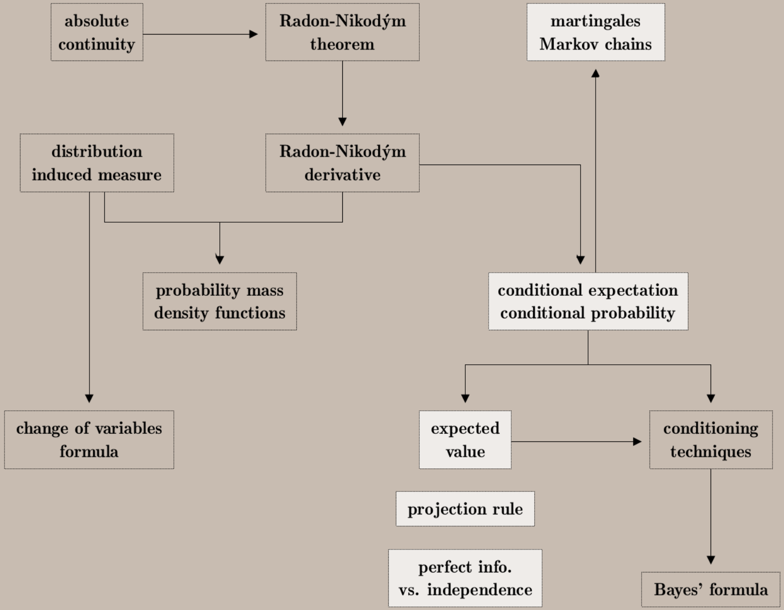

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Projection rule: $\mathscr{H} \subset \mathscr{G} \subset \mathscr{F} \implies E(E(X \,|\, \mathscr{G}) \,|\, \mathscr{H}) = E(E(X \,|\, \mathscr{H}) \,|\, \mathscr{G}) = E(X \,|\, \mathscr{H})$.

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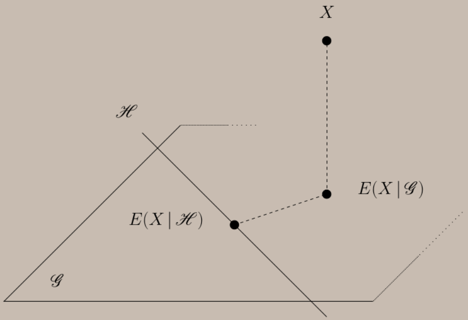

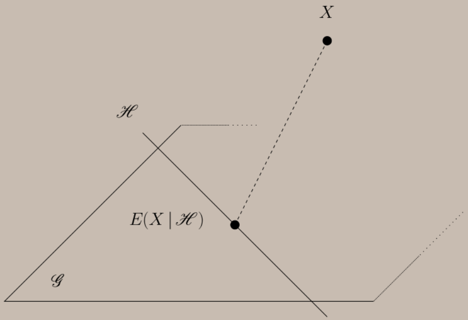

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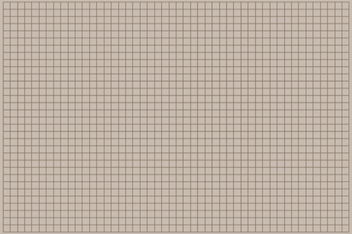

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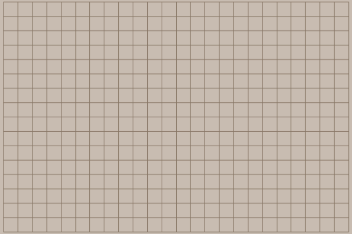

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Conditional expectation $E(X \mid \mathscr{H})$

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$$1)\ Z \in \mathscr{G} \quad \text{and} \quad 2)\ E(X\mathbf{1}_A) = E(Z\mathbf{1}_A) \quad \text{for all} \quad A \in \mathscr{G}.$$

Take $A = \Omega$ in the definition $\implies E(E(X \,|\, \mathscr{G})) = E(X)$.

Projection rule: $\mathscr{H} \subset \mathscr{G} \subset \mathscr{F} \implies E(E(X \,|\, \mathscr{G}) \,|\, \mathscr{H}) = E(E(X \,|\, \mathscr{H}) \,|\, \mathscr{G}) = E(X \,|\, \mathscr{H})$.

Perfect information: $X \in \mathscr{G} \implies E(X \,|\, \mathscr{G}) = X$.

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Independence: $Y \perp \mathscr{G}$, i.e., $P(A \cap B) = P(A)P(B) \ \ \forall A \in \sigma(Y), B \in \mathscr{G} \implies E(Y \,|\, \mathscr{G}) = E(Y)$.

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Proof. $Z = E(Y)$ is constant so $Z \in \mathscr{G}$.

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Proof. $Z = E(Y)$ is constant so $Z \in \mathscr{G}$.

Because Y and $\mathbf{1}_A$ are independent for all $A \in \mathscr{G}$,

$$E(Y\mathbf{1}_A) \overset{\text{IND}}{=} E(Y)E(\mathbf{1}_A) \overset{\text{LIN}}{=} E(E(Y)\,\mathbf{1}_A) = E(Z\mathbf{1}_A).$$

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Sum and product: $X \in \mathscr{G}$ and $Y \perp \mathscr{G} \implies$

$$E(X + Y \,|\, \mathscr{G}) \overset{\text{LIN}}{=} X + E(Y) \quad \text{and} \quad E(XY \,|\, \mathscr{G}) = X E(Y).$$

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Proof. Assume $X = \mathbf{1}_B$ for some $B \in \mathscr{G}$. Because $A \cap B \in \mathscr{G}$,

$$E(XY\mathbf{1}_A) = E(Y\mathbf{1}_{A \cap B}) = E(E(Y \,|\, \mathscr{G})\,\mathbf{1}_{A \cap B}) = E(E(Y)\,\mathbf{1}_{A \cap B}) = E(X E(Y)\,\mathbf{1}_A).$$

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Use linearity + monotone convergence to deal with X simple $\to$ positive $\to$ integrable.

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Random sum: $Y_1, Y_2, \ldots, Y_n, \ldots$ identically distributed and $\perp X : \Omega \to \mathbb{N} \implies$

$$E(Y_1 + Y_2 + \cdots + Y_X \,|\, X) = X E(Y_1).$$

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Proof. It suffices to prove 2) for the event $A = A_n = \{X = n\}$.

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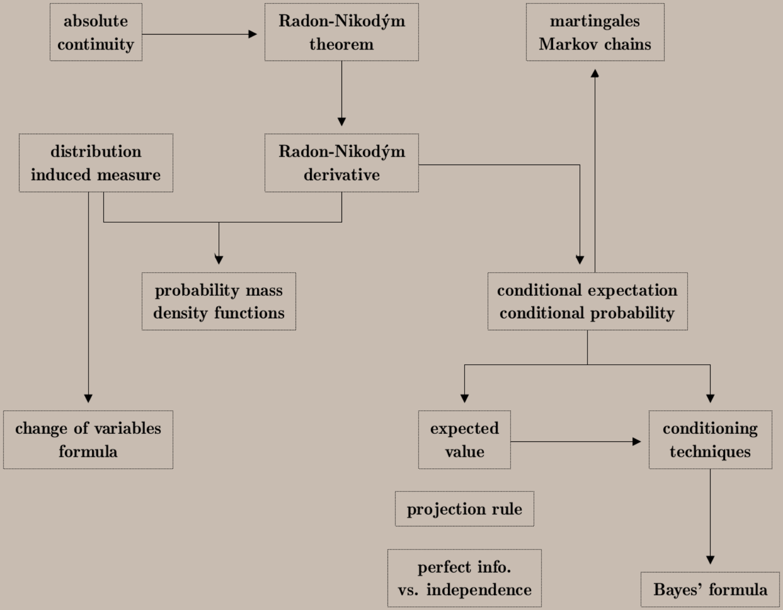
Proof. It suffices to prove 2) for the event $A = A_n = \{X = n\}$.

$$\begin{aligned} E((Y_1 + Y_2 + \cdots + Y_X)\, \mathbf{1}_{An}) &= E((Y_1 + Y_2 + \cdots + Y_n)\, \mathbf{1}_{An}) \\ &= E(nY_1 \mathbf{1}_{A_n}) = E(nE(Y_1) \mathbf{1}_{A_n}) = E(XE(Y_1) \mathbf{1}_{A_n}). \end{aligned}$$



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